

RISHAB RAJ

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PROFESSIONAL SUMMARY

Final-year Civil Engineering student at NIT Raipur with hands-on ML and full-stack experience across transportation demand modeling, computer vision, and clinical decision-support systems. INSA Summer Research Fellow (IIT Bombay) — built a hybrid VAE + IPU population synthesis pipeline for transportation microsimulation. Shipped two full-stack ML products end-to-end, from data pipeline and model training to API deployment. Amazon ML Summer School 2026 Scholar; SIH national finalist.

EDUCATION

B.Tech in Civil Engineering — National Institute of Technology, Raipur	Aug 2024 - Present
Central Board of Secondary Education (Class XII) — Shanti Niketan Academy, Gaya, Bihar	2022 - 2023
Central Board of Secondary Education (Class X) — D.A.V. Public School, Cantt. Area, Gaya, Bihar	2020 - 2021

EXPERIENCE

Indian National Science Academy (INSA) Research Fellow — IIT Bombay	May 2026 - Jul 2026
- Built a hybrid Population Synthesis pipeline integrating Variational Autoencoders (VAE) with Iterative Proportional Updating (IPU) to generate realistic census-scale synthetic populations from limited survey data. - Generated diverse synthetic population profiles using a VAE and recalibrated them with IPU to preserve demographic relationships while ensuring consistency with official census statistics for agent-based transportation microsimulation.	

PERSONAL PROJECTS

FinCA — AI-Powered Personal Financial Advisor Platform	Jul 2026
Full-stack financial planning platform that generates personalized investment plans using ML-driven risk profiling and Monte Carlo simulation.	
- Trained and benchmarked 6 ML models (Logistic Regression, SVM, Random Forest, XGBoost, LightGBM, K-Means) on 510 synthetic financial profiles to classify users into aggressive/moderate/conservative risk categories, selecting the best model via 5-fold cross-validation - Built a fully vectorized Monte Carlo engine (NumPy) running 800 parallel SIP simulations using real Nifty volatility data to project worst-case, likely, and best-case retirement corpus outcomes - Engineered a financial planning module computing income surplus, risk-based asset allocation, weighted portfolio returns, corpus projection, and Indian income-tax regime calculations, backed by a SQLite database of live Nifty, gold, FD, and inflation rates - Deployed a Flask REST API (with CORS) serving a connected JavaScript/HTML frontend, exposing endpoints for market data, active ML model info, and full financial plan calculation	
Embrya — Embryo Viability Analysis Platform	Dec 2025
Full-stack clinical-decision-support platform that scores IVF embryo viability from microscope images.	
- Developed an ensemble machine learning pipeline of three SMOTE-balanced Random Forest classifiers to predict embryo viability from extracted morphological features. - Built an OpenCV-based feature extraction engine computing circularity, symmetry, fragmentation, edge density, and entropy from embryo images for model inference. - Engineered a FastAPI backend exposing a REST prediction endpoint with request validation, feature scaling, and probability-averaged ensemble inference. - Developed a React + TypeScript clinical dashboard (Tailwind CSS, Radix UI) with real-time scoring, batch processing, embryo comparison, and PDF/TXT report generation.	

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, JavaScript, HTML, CSS
Data Analytics: Data Analysis, Microsoft Excel, Pandas
Machine Learning & Backend: Machine Learning, FastAPI, REST API Development
Developer Tools: VS Code, Jupyter Notebook
Soft Skills: Analytical Thinking, Problem Solving, Process Optimization, Teamwork, Leadership, Communication
Areas of Interest: Artificial Intelligence, Data Analytics, Transportation Engineering

POSITIONS OF RESPONSIBILITY

Executive, Civil Engineering Association (CEA), NIT Raipur	Jul 2025 - Present
Coordinator, National Service Scheme (NSS), NIT Raipur	Oct 2024 - Present

ACHIEVEMENTS

Amazon ML Summer School 2026 Scholar, Amazon	Jul 2026
INSA Summer Research Fellow, Indian National Science Academy (INSA)	May 2026
National Finalist, Smart India Hackathon (SIH)	Dec 2025
National Level Merit Holder, Indian Olympiad Qualifier in Mathematics (IOQM)	Sept 2020
National Level Merit Holder, AIR 15, Indian National Mathematics Talent Contest	Nov 2018